Technical Report: Green-Ops Profiler

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Abstract

This report details the development and deployment of the Green-Ops Profiler, a containerized telemetry suite designed to audit the carbon footprint and energy efficiency of deep learning models. By implementing hardware-aware monitoring and standardized benchmarking protocols, we establish a quantitative "Green Frontier" for architectural selection in energy-constrained environments.

1 Executive Summary

The Green-Ops Profiler addresses the opacity of AI energy consumption during the inference phase. This project engineered a modular system capable of capturing real-time hardware metrics, including Energy (Wh), Latency (s), and Carbon Emissions (kg CO_2eq). The framework enables developers to transition from accuracy-only evaluation to a multi-dimensional sustainability audit.

2 System Architecture

The system is built upon a three-layer decoupled architecture:

1. **Infrastructure Layer (Docker):** Provides environment parity and ensures that hardware monitoring tools are reproducible across different host OS environments.
2. **Telemetry Layer (CodeCarbon/NVML):** Interfaces directly with Intel RAPL and NVIDIA NVML sensors to capture high-resolution power draw data.
3. **Application Layer (PyTorch):** Manages model lifecycles and standardizes input tensors for consistent benchmarking.

3 Methodology

To ensure scientific rigor and eliminate hardware-induced bias, a strict execution protocol was followed:

- **Thermal Warm-up:** Each model undergoes 10 non-measured warm-up iterations to stabilize hardware clock speeds and thermal states.
- **Resource Isolation:** All benchmarks are executed using `torch.no_grad()` to prevent gradient overhead. Models are explicitly purged from memory (`del`) and the CUDA cache is cleared between sessions.
- **Temporal Filtering:** The reporting logic isolates data based on the session start time, allowing for a persistent historical log (`emissions.csv`) while generating clean, experiment-specific summaries.

4 Experimental Results

The following table summarizes the benchmarking results for various vision architectures on CPU-based infrastructure.

Table 1: Comparative Environmental Metrics (Session Data)

Model Label	**Avg. Latency (s)**	**Energy (Wh)**	**CO_2 (kg)**	**Efficiency Score**
MobileNet-V2	5.70	0.020	0.000009	0.114
ResNet-18	5.89	0.019	0.000009	0.112
ResNet-50	13.46	0.022	0.000011	0.296
ViT-B-16	29.06	0.042	0.000020	1.220

Efficiency Score : $Energy \times Latency$ (Lower is Better).

5 Discussion and Technical Insights

The empirical data reveals a non-linear relationship between architectural complexity and energy consumption.

5.1 Diminishing Returns of Depth

While ResNet-50 provides superior feature extraction, the 128% increase in latency compared to ResNet-18 suggests that for real-time edge applications, the energy cost outweighs the incremental gain in precision.

The **ViT-B-16** (Vision Transformer) represents the highest environmental cost in the suite. With a latency of 29.06s, it is nearly **5 times slower** than the ResNet-18 baseline and consumes **double the energy** per session. The efficiency score of 1.220 (where lower is better) confirms that without specific optimization, Transformers impose a heavy "sustainability tax" that may be prohibitive for real-time edge applications.

5.2 Calculated Efficiency Score

The `efficiency_score` ($Energy \times Latency$) effectively penalizes models that are both slow and energy-intensive. The exponential jump in this score for the ViT-B-16 architecture highlights the critical need for the quantization and pruning strategies.

6 Conclusion

This project has successfully established a high-precision, reproducible instrument for AI energy auditing. This foundation allows for the systematic identification of “energy-intensive” layers, enabling data-driven optimization strategies.